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Non-electrolytic capacitors

in single-phase split motors. Connecting a capacitor of suitable value in series with the auxiliary winding, which is in parallel to the main coil, results in phase splitting. Single-phase power is converted into two-phase, which provides the starting torque.

Snubber circuit. Triacs are susceptible to 'rate effect' problem. Internal capacitance exists between the main terminals and gate of a triac. If a sharply rising voltage appears even on the main terminal, it can cause enough break through to the gate to trigger the triac 'on'. This unwanted 'rate effect' turn-on can be caused by supply-line transients.

The problem is particularly severe when driving inductive loads such as induction motors. It is overcome by a snubber circuit comprising a resistor in series with a capacitor, which is connected between MT1 and MT2 terminals of the triac. It is a safety arrangement against false triggering.

Q2. What is the main difference between VHDL and Verilog HDL? Which one is widely used in the industry for designing ICs?

Vishal Kulkarni

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A2. Verilog and VHDL are hardware description languages used to write programs for electronic chips. These languages are used in electronic devices that do not share a computer's basic architecture. VHDL is the older of the two, and is based on Ada and Pascal, thus inheriting characteristics from both languages. Verilog is relatively recent and follows the coding methods of C programming language, which makes it easier to grasp

and understand.

VHDL is a strongly-typed language that does not allow intermixing or operation of variables with different classes. On the other hand, Verilog uses weak typing.

VHDL is case-insensitive and users can freely change the case as long as the characters in the name and the order stay the same. Verilog is case-sensitive and doesn't recognise a variable if the case used is inconsistent with the previous one.

VHDL has the advantage of having a lot more constructs that aid in high-level modeling, and it reflects the actual operation of the device being programmed. Complex data types and packages are very desirable when programming big and complex systems that might have a lot of functional parts. Verilog has no concept of packages and all programming must be done with the simple data types that are provided by the programmer.

In VHDL, a library is a store for compiled entities, architectures, packages and configurations useful for managing multiple design projects.

Capability-wise, hardware structure can be modeled equally effectively in both VHDL and Verilog. When modeling abstract hardware, the capability of VHDL can sometimes only be achieved in Verilog when using the programming language interface. The choice of which language to use is not therefore based solely on technical capability but on personal preferences as well.

Verilog is the language of choice for industrial applications that require both simulation and synthesis. It lacks, however, constructs needed for system-level specifications. VHDL is more complex, thus difficult to learn and use. However, it offers a lot more flexibility of the coding styles and is suitable for handling very complex designs.

Answers compiled by Col (Retd) N.C. Pande.

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